

Column-wise Mixed Noise Perturbation

1 Data notation

Let $D \in \{S, T\}$ denote the source or target domain, let $u \in \mathcal{U}_D$ denote a user, and let $i \in \{1, 2, 3\}$ denote a data modality. At time t_ν , the observed data matrix for modality i is written as

$$Z_{i,D}^u(t_\nu) \in \mathbb{R}^{N_d \times m_i}, \quad \nu = 0, \dots, N_t.$$

Here N_d is the number of days and m_i is the number of features in modality i . In the numerical experiments,

$$(m_1, m_2, m_3) = (14, 17, 30),$$

corresponding to smartwatch, intelligent-insole, and EMR features, respectively. The r -th row of $Z_{i,D}^u(t_\nu)$ is the observation on day r , and the k -th column is feature k .

For an evaluation day d_j^D , the row vector used by the model is

$$B_{i,D}^{u,j}(t_\nu) := (Z_{i,D}^u(t_\nu))_{[d_j^D, :] }.$$

Thus $Z_{i,D}^u(t_\nu)$ is the day-by-feature data matrix at time t_ν , whereas $B_{i,D}^{u,j}(t_\nu)$ is the single evaluation-day row extracted from $Z_{i,D}^u(t_\nu)$.

To define noise at the column level, fix a user u , modality i , and feature column k . Collect the complete day-time matrix for that feature as

$$X_{r,\nu} = (Z_{i,D}^u(t_\nu))_{r,k}, \quad r = 1, \dots, N_d, \quad \nu = 0, \dots, N_t.$$

Thus $X \in \mathbb{R}^{N_d \times (N_t+1)}$ is one feature matrix extracted from the larger data object $Z_{i,D}^u$. In the experiments reported in the paper, noise is applied to all days, so the scale of a noisy feature is computed from the pooled empirical standard deviation over all entries of this day-time matrix:

$$\tau_X := \text{sd}\{X_{r,\nu} : r = 1, \dots, N_d, \nu = 0, \dots, N_t\}.$$

The implementation also supports applying noise only to selected day sets $\mathcal{R}$, such as held-out test days or reference days. In that case, the same definition is used with $r \in \mathcal{R}$ in place of all days.

2 Elementary perturbation mechanisms

Let $s > 0$ be a dimensionless perturbation level. Under the medium calibration used in the experiments, define

$$\sigma_{\text{add}} = 0.025s, \quad \sigma_{\text{drift}} = 0.015s, \quad \sigma_{\text{mult}} = 0.030s, \quad \sigma_{\text{cum}} = 0.040s.$$

For a feature matrix X , the elementary perturbation mechanisms are defined row-wise in time, with the scale shared across the selected rows of the same user and feature.

Additive noise.

$$\tilde{X}_{r,\nu} = X_{r,\nu} + \epsilon_{r,\nu}, \quad \epsilon_{r,\nu} \sim N(0, \sigma_{\text{add}}^2 \tau_X^2).$$

Drift noise.

$$\tilde{X}_{r,\nu} = X_{r,\nu} + d_{r,\nu}.$$

For each selected day r , $d_r = (d_{r,0}, \dots, d_{r,N_t})$ is generated from a random walk, centered over time, and rescaled so that

$$\text{sd}(d_r) = \sigma_{\text{drift}} \tau_X.$$

Multiplicative noise.

$$\tilde{X}_{r,\nu} = X_{r,\nu}(1 + \epsilon_{r,\nu}), \quad \epsilon_{r,\nu} \sim N(0, \sigma_{\text{mult}}^2).$$

Cumulative noise. Let

$$\Delta X_{r,\ell} = X_{r,\ell} - X_{r,\ell-1}, \quad \ell = 1, \dots, N_t,$$

and let

$$\tau_{\Delta X} := \text{sd}\{\Delta X_{r,\ell} : r = 1, \dots, N_d, \ell = 1, \dots, N_t\},$$

again restricted to $r \in \mathcal{R}$ if only selected days are perturbed. The cumulative perturbation is applied at the increment level and then reconstructed by summation:

$$\tilde{X}_{r,0} = X_{r,0}, \quad \tilde{X}_{r,\nu} = X_{r,0} + \sum_{\ell=1}^{\nu} [\Delta X_{r,\ell} + \eta_{r,\ell}]_+,$$

where

$$\eta_{r,\ell} \sim N(0, \sigma_{\text{cum}}^2 \tau_{\Delta X}^2), \quad [a]_+ = \max\{a, 0\}.$$

The positive-part operator enforces nonnegative increments for cumulative-count variables.

3 Mixed noise operator

The mixed perturbation operator $\mathcal{M}_{\text{mix}}$ is a column-wise and type-aware operator. It does not apply all elementary noises to every feature. Instead, each feature column is assigned the perturbation mechanism that matches its data type:

$$\tilde{Z}_{i,D}^u(t_\nu) = \mathcal{M}_{\text{mix}}(Z_{i,D}^u)(t_\nu), \quad \nu = 0, \dots, N_t.$$

After perturbing $Z_{i,D}^u$, the model receives the corresponding evaluation row

$$\tilde{B}_{i,D}^{u,j}(t_\nu) = (\tilde{Z}_{i,D}^u(t_\nu))_{[d_j^p, :] }.$$

The mixed perturbation is specified by the following feature-to-noise assignment. In the formulas below, $\Pi_{\mathcal{I}}$ denotes clipping to the feasible range of a feature, such as nonnegativity or a percentage interval.

Feature group	Noise in mixed mode	Formula applied to each feature matrix X
Smartwatch continuous non-cumulative features: heart rate, oxygen saturation, calories, skin temperature, HRV, stress index, blood pressure, activity intensity	Additive plus drift	$\tilde{X}_{r,\nu} = \Pi_{\mathcal{I}}(X_{r,\nu} + \epsilon_{r,\nu} + d_{r,\nu}),$ $\epsilon_{r,\nu} \sim N(0, \sigma_{\text{add}}^2 \tau_X^2).$
Smartwatch cumulative-count features: steps, active minutes, floors climbed	Cumulative increment noise	$\tilde{X}_{r,0} = X_{r,0},$ $\tilde{X}_{r,\nu} = X_{r,0} + \sum_{\ell=1}^{\nu} [\Delta X_{r,\ell} + \eta_{r,\ell}]_+,$ $\eta_{r,\ell} \sim N(0, \sigma_{\text{cum}}^2 \tau_{\Delta X}^2).$
Smartwatch discrete features: sleep quality, activity category	No perturbation	$\tilde{X}_{r,\nu} = X_{r,\nu}.$
Intelligent-insole main continuous features: plantar pressure, step frequency, stride length, contact time, force estimate, exercise intensity, plantar fatigue	Multiplicative noise	$\tilde{X}_{r,\nu} = \Pi_{\mathcal{I}}(X_{r,\nu}(1 + \epsilon_{r,\nu})),$ $\epsilon_{r,\nu} \sim N(0, \sigma_{\text{mult}}^2).$
Intelligent-insole ratio and score features: symmetry, balance score, left-right ratio	Additive plus drift	$\tilde{X}_{r,\nu} = \Pi_{\mathcal{I}}(X_{r,\nu} + \epsilon_{r,\nu} + d_{r,\nu}),$ $\epsilon_{r,\nu} \sim N(0, \sigma_{\text{add}}^2 \tau_X^2).$
Intelligent-insole center-coordinate features: center x , center y	Additive plus drift	$\tilde{X}_{r,\nu} = X_{r,\nu} + \epsilon_{r,\nu} + d_{r,\nu},$ $\epsilon_{r,\nu} \sim N(0, \sigma_{\text{add}}^2 \tau_X^2).$
Intelligent-insole discrete features: gait type, motion type	No perturbation	$\tilde{X}_{r,\nu} = X_{r,\nu}.$
Static EMR features	No perturbation	$\tilde{Z}_{3,D}^u(t_\nu) = Z_{3,D}^u(t_\nu).$

In short, mixed noise is not a direct sum of additive, drift, multiplicative, and cumulative noise on every feature. It is a feature-wise assignment rule: cumulative variables receive cumulative noise, continuous smartwatch variables receive additive-plus-drift noise, main continuous insole variables receive multiplicative noise, ratio and score insole variables receive clipped additive-plus-drift noise, center-coordinate insole variables receive unclipped additive-plus-drift noise, and discrete or static clinical variables are left unchanged.